Liem Fairbairn

+44 (0)7859 951239 | lbfairbairn@gmail.com | LinkedIn

EDUCATION

Imperial College London — MEng Aeronautical Engineering (4-year Integrated Master's) 2025 – 2029
Year 1 Cumulative GPA: 3.6/4 (US equivalent)

Alexandra Park School 2018 – 2025
A Levels: Mathematics (A*), Physics (A*), Classical Civilisation (A*), Further Mathematics (A)

TECHNICAL SKILLS

Python, MATLAB, Dassault 3DEXPERIENCE (Part Design, Assembly, Drafting), STAR-CCM+ (CFD), Abaqus (FEA), Experimental Testing & Data Analysis, Technical Report Writing

FEATURED PROJECT

NASA TechLeap Robotically Manipulated Payload Challenge — Independent Submission 2026

- Independently authored a full engineering proposal for an in-space xenon propellant transfer payload, directly addressing a named NASA electric propulsion capability gap (Shortfall 23.07).
- Sized dual pressure vessels (6061-T6 aluminium, 42 MPa worst-case design pressure), and modelled supercritical blowdown dynamics using a real-gas Redlich-Kwong equation of state in MATLAB
- Designed a motorised valve control system and electromagnetic docking/locking mechanism for autonomous fluid transfer via a Free-Flying Robot arm, with an Arduino-based sensor and data-logging.
- Produced CAD concept assembly (Dassault 3DEXPERIENCE), cost/budget narrative, engineering drawings, and video pitch as sole engineer on the submission.

ENGINEERING PROJECTS

2 Blade Wind Turbine Design and Manufacturing 2026

- Designed and manufactured two turbine blades in a team of 6, extracting 13.9 W in a 10 m/s wind tunnel at 1600 rpm; developed a MATLAB Blade Element Momentum solver with aerodynamic correction models to predict performance.

Structural, Materials & Aerodynamics Laboratory Work 2025 – 2026

- Characterised elastic and plastic material response via tensile testing, and validated pin-joint and beam-loading experiments against FEA predictions.
- Derived friction coefficient–Reynolds number relationships from internal pipe flow testing, and measured lift, drag, and pitching moment on a model jet across varying tailplane deflection.

Pressure Transducer Node Analysis — Numerical Methods 2026

- Developed MATLAB numerical solvers for Bessel functions to identify diaphragm vibration modes within a pressure transducer model.

Urban Electric Vehicle Design 2025

- Designed a climate-focused urban vehicle in a 6-person team, producing CAD models and technical drawings in Dassault 3DEXPERIENCE.

LEADERSHIP & COMPETITIONS

M3 Mathematics Modelling Challenge 2025

- Led a seven-person team, developing Python-based simulations of thermal behaviour in buildings and infrastructure resilience under power outage scenarios.

ISSET Mission Discovery Project 2024

- Collaborated in a team of 6 to design a microgravity-centred experiment assessing changes in the material properties of string-like materials.

Astrophysics Educator — Alexandra Park Sixth Form 2022 – 2025

- Founded an Astrophysics and Engineering club, delivering weekly lectures to 10–15 students on GCSE Astronomy and related topics, building strong communication and leadership skills.